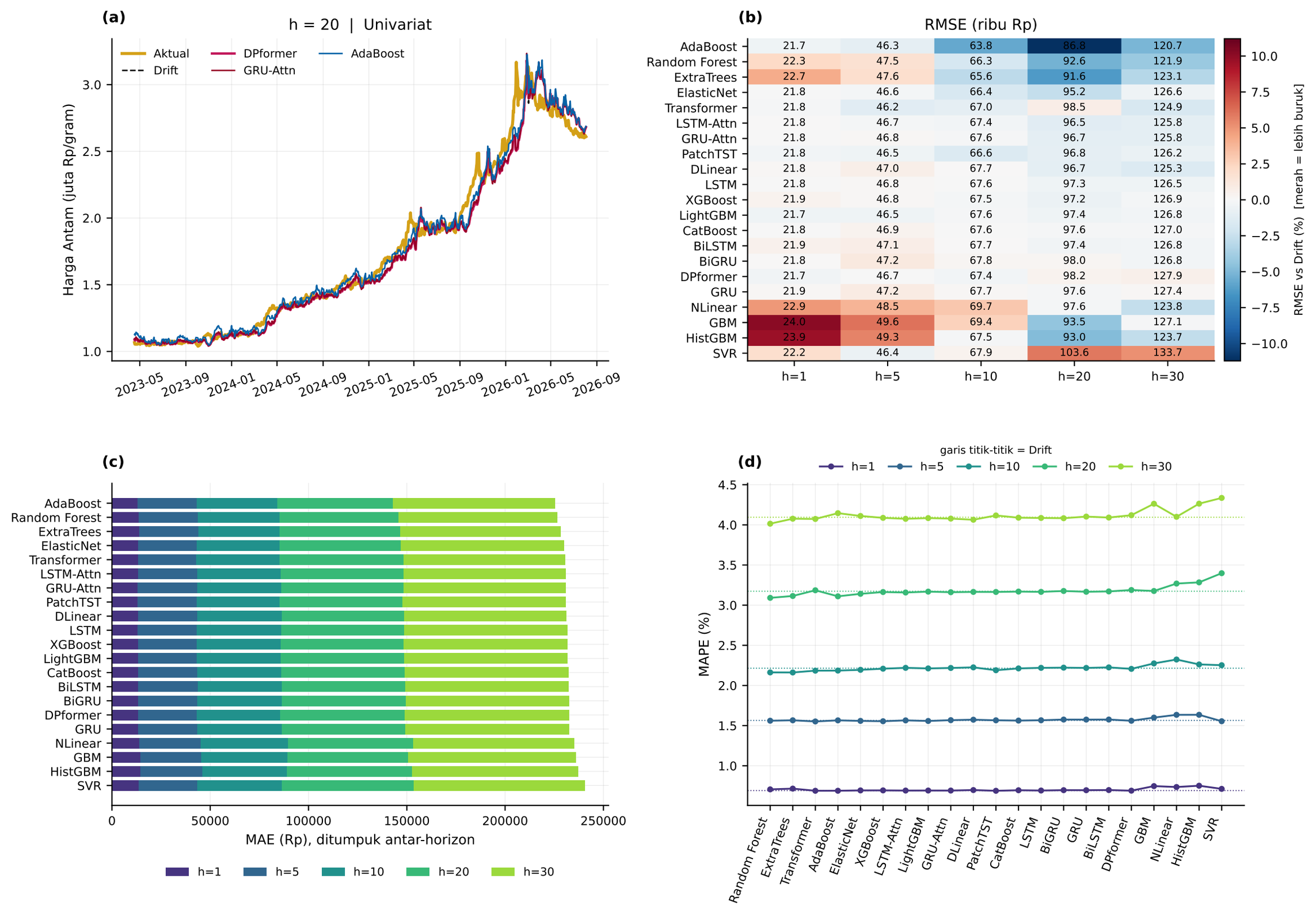

(a)
h = 20 | Univariat
Aktual
Drift
DPformer
GRU-Attn
AdaBoost
Harga Antam (juta Rp/gram)
3.0
2.5
2.0
1.5
1.0
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
2026-05
2026-09
(b)
RMSE (ribu Rp)
AdaBoost 21.7 46.3 63.8 86.8 120.7
Random Forest 22.3 47.5 66.3 92.6 121.9
ExtraTrees 22.7 47.6 65.6 91.6 123.1
ElasticNet 21.8 46.6 66.4 95.2 126.6
Transformer 21.8 46.2 67.0 98.5 124.9
LSTM-Attn 21.8 46.7 67.4 96.5 125.8
GRU-Attn 21.8 46.8 67.6 96.7 125.8
PatchTST 21.8 46.5 66.6 96.8 126.2
DLinear 21.8 47.0 67.7 96.7 125.3
LSTM 21.8 46.8 67.6 97.3 126.5
XGBoost 21.9 46.8 67.5 97.2 126.9
LightGBM 21.7 46.5 67.6 97.4 126.8
CatBoost 21.8 46.9 67.6 97.6 127.0
BiLSTM 21.9 47.1 67.7 97.4 126.8
BiGRU 21.8 47.2 67.8 98.0 126.8
DPformer 21.7 46.7 67.4 98.2 127.9
GRU 21.9 47.2 67.7 97.6 127.4
NLinear 22.9 48.5 69.7 97.6 123.8
GBM 24.0 49.6 69.4 93.5 127.1
HistGBM 23.9 49.3 67.5 93.0 123.7
SVR 22.2 46.4 67.9 103.6 133.7
h=1
h=5
h=10
h=20
h=30
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
RMSE vs Drift (%) [merah = lebih buruk]
(c)
AdaBoost
Random Forest
ExtraTrees
ElasticNet
Transformer
LSTM-Attn
GRU-Attn
PatchTST
DLinear
LSTM
XGBoost
LightGBM
CatBoost
BiLSTM
BiGRU
DPformer
GRU
NLinear
GBM
HistGBM
SVR
0
50000
100000
150000
200000
250000
MAE (Rp), ditumpuk antar-horizon
h=1
h=5
h=10
h=20
h=30
(d)
garis titik-titik = Drift
h=1
h=5
h=10
h=20
h=30
MAPE (%)
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
Random Forest
ExtraTrees
Transformer
AdaBoost
ElasticNet
XGBoost
LSTM-Attn
LightGBM
GRU-Attn
DLinear
PatchTST
CatBoost
LSTM
BiGRU
GRU
BiLSTM
DPformer
GBM
NLinear
HistGBM
SVR